

Ticket 05 Study Report: American Put Validation

SURF2026 American Option Risk Surfaces

Codex-assisted implementation report

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Abstract

Ticket 05 uses the Ticket 04 American CN/PSOR solver to validate American put behavior in a representative Black-Scholes-with-dividends case. The ticket produces validation tables, selected spot checks, obstacle and convergence summaries, and a value-versus-payoff figure. This is a validation ticket, not a new solver ticket. It does not implement boundary extraction, Greeks, stress maps, datasets, neural-network files, or American call validation.

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1 Role of Ticket 05 in the Whole Solver Project

Tickets 01 through 04 built the solver foundation in layers. Ticket 01 implemented payoff and European Black-Scholes utilities. Ticket 02 implemented finite-difference grids and the Black-Scholes spatial operator. Ticket 03 validated European Crank-Nicolson time stepping against closed-form European prices. Ticket 04 added the PSOR/LCP core needed for the American payoff obstacle.

Ticket 05 is the first validation ticket for a genuinely American option case. It does not add a new pricing algorithm. Instead, it uses the Ticket 04 American CN/PSOR solver and asks whether the American put outputs look financially and numerically credible in a representative setup.

The role of this ticket is evidence gathering:

- check that computed American put values remain above payoff;
- check that computed American put values are at least European put values;
- summarize PSOR convergence metadata;
- inspect selected spot prices in the main region of interest;
- compare medium and fine grids lightly, without claiming full grid sensitivity.

This ticket intentionally stops before free-boundary extraction. It may report the obstacle gap $U - \Phi$, because that is a validation diagnostic. It does not turn that gap into a boundary location.

2 Theory and Methodology Connection

The documented source basis for Ticket 05 is internal:

- `reports/01_literature/literature_map.md`;
- `reports/02_math/formulation_note.md`;
- `reports/03_solver/solver_validation_plan.md`;
- `reports/03_solver/tickets/ticket_04_psor_lcp_core.tex`.

The literature map identifies American option pricing as an obstacle problem. The formulation note expresses the American option conditions as a variational inequality and then as a linear complementarity problem after discretization. The solver validation plan states that American put validation should include obstacle checks, continuation-premium or obstacle-gap sanity checks, selected prices, and convergence evidence.

This reference integration pass adds verified citations for the external methodology behind the internal source basis. American put finite-difference validation is connected to Brennan-Schwartz (Brennan and Schwartz, 1977); the LCP and obstacle language is connected to standard LCP methodology (Cottle et al., 2009); and the European comparison benchmark rests on the Black-Scholes/Merton framework (Black and Scholes, 1973; Merton, 1973).

3 Why American Put Validation Follows the PSOR/LCP Core

A European validation run checks the continuation PDE. An American put validation run checks the continuation PDE plus the early-exercise constraint. The American constraint is

$$U(S, \tau) \geq \Phi_{\text{put}}(S), \quad \Phi_{\text{put}}(S) = \max(K - S, 0). \quad (1)$$

Ticket 04 built the PSOR machinery that enforces this constraint at each Crank-Nicolson step. Ticket 05 comes next because the solver core must now be used in a realistic put case, with reported evidence rather than only smoke tests.

For beginner researchers, the distinction is important:

- Ticket 04 asks: can the code enforce the obstacle in a reusable solver core?
- Ticket 05 asks: when that core is used for an American put, do the prices and diagnostics obey the basic financial checks?

4 American Put Financial Intuition

An American put gives the holder the right to sell the underlying asset for strike K at any time up to maturity. When the spot S is low, the payoff $K - S$ is high. Early exercise can therefore be attractive for puts, especially when the put is deep in the money.

The American put should be at least as valuable as:

- immediate exercise payoff, because the holder can exercise now;
- the European put, because the American holder has every European exercise choice plus the extra right to exercise early.

These two inequalities are not a complete proof of pricing accuracy, but they are essential financial sanity checks. If either inequality fails materially, the American solver should not be trusted for later boundary, Greek, stress-map, dataset, or neural-surrogate work.

5 Validation Criteria

5.1 Obstacle Condition

For the American put, the value grid should satisfy

$$U_{n,i} \geq \Phi_i \quad (2)$$

at every tested time and spot node. Ticket 05 reports

$$\min_{n,i} (U_{n,i} - \Phi_i), \quad (3)$$

$$\max_{n,i} \max(\Phi_i - U_{n,i}, 0). \quad (4)$$

The first quantity is the minimum obstacle gap. The second is the maximum obstacle violation.

5.2 American Versus European Put

The American put should also satisfy

$$U^{\text{American}}(S, T) \geq P^{\text{European}}(S, T) \quad (5)$$

up to numerical tolerance. Ticket 05 compares final American values against the Ticket 01 European Black-Scholes put utility.

5.3 Convergence Metadata

Each PSOR time step returns metadata. Ticket 05 reports whether all steps converged, how many PSOR iterations were needed, and the maximum final update. This matters because an obstacle check alone can miss a solver that stopped too early.

5.4 Selected Spots and Medium-to-Fine Comparison

Selected spot checks use S/K values 0.5, 0.8, 1.0, 1.2, and 1.5. These make the table readable for students because each row corresponds to a familiar moneyness point. The medium-to-fine comparison is intentionally light: it compares selected prices from $M = N = 120$ and $M = N = 180$. Full grid and domain sensitivity belongs to a later ticket.

6 Obstacle Gap Is Not Boundary Extraction

Ticket 05 reports the obstacle gap

$$G(S, \tau) = U(S, \tau) - \Phi(S). \quad (6)$$

This gap is useful because it directly checks the obstacle condition. If $G < 0$, the solver has violated the American payoff constraint. If $G = 0$ at some grid points, those points may be consistent with immediate exercise.

However, Ticket 05 does not extract a free boundary. Boundary extraction requires option-type specific search direction, thresholds, interpolation, and missing-boundary metadata. That is planned for a later ticket. In this report, obstacle gap means validation evidence only.

7 Implementation Design Decisions

Decision	Reason
Use Ticket 04 solver as-is	Ticket 05 is a validation layer. Solver changes are out of scope unless a genuine bug is found.
Use $K = 1$, $S_{\max} = 4$, $T = 1$, $r = 0.05$, $q = 0.02$, $\sigma = 0.20$	These values match earlier ticket defaults and the validation plan's normalized-strike workflow.
Use medium and fine grids	This gives a preliminary refinement check without turning Ticket 05 into full grid/domain sensitivity.

Decision	Reason
Write CSV artifacts	Tables are easy to inspect, test, and reuse in reports.
Create one figure	An American put validation ticket benefits from one visual comparison of American value, payoff, and European value.
Avoid boundary extraction language	Obstacle gap reporting is allowed, but boundary location inference is explicitly out of scope.

8 Code-Level Function Explanations

Function	Purpose
<code>run_validation_cases</code>	Runs the medium and fine American put cases, returns validation rows, selected-spot rows, and artifact metadata.
<code>summarize_american_put_case</code>	Builds one obstacle, European-comparison, and PSOR convergence summary row.
<code>selected_spot_rows</code>	Reports payoff, European put, American put, American-minus-European, and American-minus-payoff at selected moneyness points.
<code>nearest_spot_index</code>	Selects the nearest grid point to a requested spot. It does not interpolate.
<code>write_csv</code>	Writes rows with stable column order.
<code>create_value_vs_payoff_figure</code>	Creates the validation figure if matplotlib is available.
<code>main</code>	Runs the experiment and writes the CSV/figure artifacts.

9 Validation Result Tables and Interpretation

9.1 Validation Cases

Case	K	T	r	q	σ	M	N
Medium	1.0	1.0	0.05	0.02	0.20	120	120
Fine	1.0	1.0	0.05	0.02	0.20	180	180

9.2 Obstacle and European Comparison Summary

Case	Min gap	Max violation	Min Am. - Eur.	Max Am. - Eur.
Medium	0	0	-3.7241×10^{-14}	0.0487706
Fine	0	0	-3.7241×10^{-14}	0.0487706

The minimum American-minus-European value is a tiny negative roundoff-scale value. It is not a financially meaningful violation. The American put value dominates the European put across the tested grid up to numerical tolerance.

9.3 Convergence Metadata

Case	All steps converged	Step count	Max iterations	Max final update
Medium	True	120	10	9.9971×10^{-9}
Fine	True	180	10	9.9919×10^{-9}

Every PSOR step converged in both validation cases. The maximum final updates are just below the requested tolerance scale, which is expected for an iterative method that stops when the update falls below tolerance.

9.4 Selected Spot Price Comparison

Case	S/K	Payoff	European put	American put	Am. - Eur.	Am. - payoff
Medium	0.5	0.500000	0.461146	0.500000	0.038854	0
Medium	0.8	0.200000	0.182378	0.200381	0.018003	0.000381
Medium	1.0	0	0.063301	0.066301	0.003001	0.066301
Medium	1.2	0	0.015602	0.015984	0.000381	0.015984
Medium	1.5	0	0.001228	0.001254	0.000026	0.001254
Fine	0.488889	0.511111	0.472032	0.511111	0.039079	0
Fine	0.8	0.200000	0.182378	0.200606	0.018228	0.000606
Fine	1.0	0	0.063301	0.066469	0.003168	0.066469
Fine	1.2	0	0.015602	0.016054	0.000451	0.016054
Fine	1.488889	0	0.001358	0.001386	0.000028	0.001386

The selected spot table shows the expected pattern: the American put is never below payoff and is above the European put. At low spot values, the American value can equal payoff exactly on the chosen grid node. This is an obstacle validation observation, not a boundary extraction claim.

9.5 Medium-to-Fine Comparison

The largest selected-spot medium-to-fine difference recorded by the experiment is 0.0111111. This value is driven by nearest-grid selection at $S/K = 0.5$: the medium grid includes $S = 0.5$, while the fine grid's nearest point is $S = 0.488889$. This is a useful reminder that selected spot tables using nearest nodes are simple and auditable, but they are not a full interpolation or convergence study.

10 Figure Interpretation

Figure 1 plots the fine-grid American put value, the payoff, and the European put value over moneyness S/K . The figure is evidence for three basic visual checks:

- the American value does not fall below payoff;
- the American value is above the European put value;
- the curves have the expected put shape, with value decreasing as spot increases.

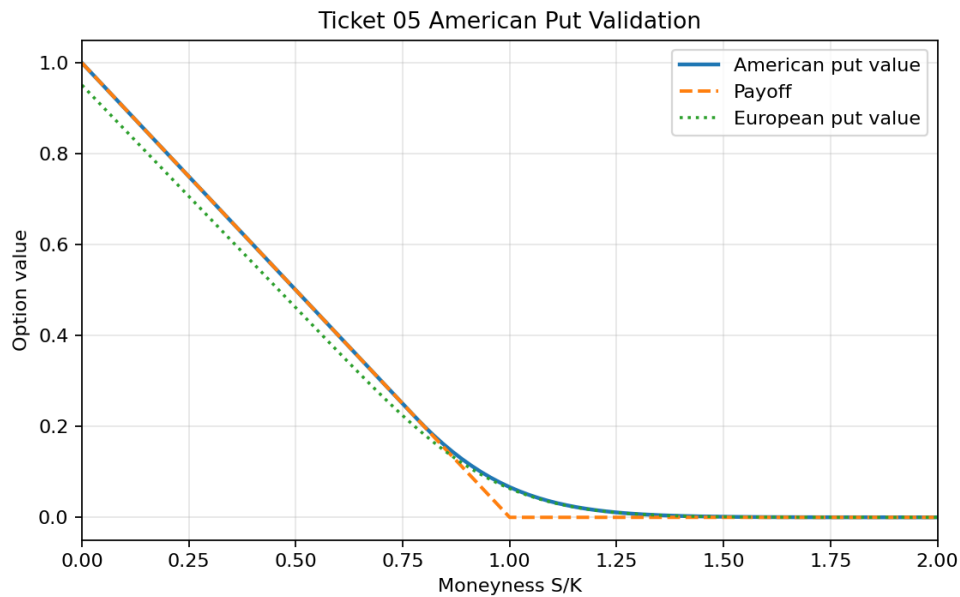


Figure 1: Ticket 05 American put value, payoff, and European put comparison. The plot is a validation visual, not a boundary-extraction plot.

11 Testing Strategy

Test category	What it proves	What it does not prove
Script/artifact test	The experiment runs and writes the expected CSV and figure files.	Full financial validation across regimes.
CSV schema tests	The result tables have stable, expected columns.	That every metric is sufficient for later tickets.
Obstacle tests	American values dominate payoff within tolerance.	Complementarity residual accuracy.
European comparison tests	American puts dominate European puts within tolerance.	External benchmark agreement for American prices.
PSOR metadata tests	Default cases converged and expose iteration/update summaries.	Convergence for all possible grids and parameters.
Nearest-grid test	Selected spot reporting is predictable.	Interpolation accuracy.
Figure test	The meaningful validation PNG is created when matplotlib is available.	That the figure alone validates the solver.
Scope tests	No boundary extraction, Greeks, stress maps, datasets, or neural APIs/files are introduced.	Future tickets that intentionally add those features.
Docstring test	New Python files identify Ticket 05.	Completeness of the research report.

12 Test Results and Interpretation

The test command was:

```
PYTHONPYCACHEPREFIX=/private/tmp/codex_pycache PYTHONPATH=src MPLCONFIGDIR=/private/
tmp/matplotlib-cache python3 -m unittest discover -s tests
```

Observed result during Ticket 05 implementation:

```
Ran 50 tests

OK
```

The experiment command was:

```
PYTHONPYCACHEPREFIX=/private/tmp/codex_pycache PYTHONPATH=src MPLCONFIGDIR=/private/
tmp/matplotlib-cache python3 experiments/02_american_put_validation.py
```

The experiment wrote two validation rows, ten selected-spot rows, and one value-versus-payoff PNG.

13 Implementation Pitfalls and Debugging Notes

Recorded item or pitfall	Symptom	Fix or guard	Beginner lesson
Expected TDD red run	The new tests failed with <code>FileNotFoundError</code> because the Ticket 05 experiment script did not exist yet.	Create <code>experiments/02_american_put_validation.py</code> .	A red test proves the tests are aimed at the new artifact.
Relative metadata paths	Two tests failed because the script returned relative artifact paths while tests expected absolute paths.	Root output paths at the project directory and return absolute metadata paths.	Tests can reveal reproducibility details, not only finance logic.
Font cache warning	Matplotlib emitted a fontconfig cache warning in the sandboxed runtime.	Use writable <code>MPLCONFIGDIR</code> and <code>FC_CACHEDIR</code> locations.	Report-generation code may need environment hygiene.
Overclaiming from one case	One representative case could be mistaken for full solver validation.	State that this is preliminary American put validation only.	Validation evidence has a scope.
Confusing obstacle gap with boundary extraction	$U - \Phi$ may tempt a reader to infer a free boundary.	Report gap values but do not infer, label, or extract a boundary.	Diagnostics and boundary extraction are different tasks.
Ignoring European comparison	Checking only $U \geq \Phi$ would miss a weak American value.	Compare against the Ticket 01 European put utility.	American value should include the European exercise right plus early exercise.
Ignoring PSOR non-convergence	Obstacle values alone could hide slow or failed PSOR steps.	Report all-step convergence, iterations, and final updates.	Numerical credibility includes solver status metadata.
Using too coarse a grid	A coarse grid can create misleading selected-spot values.	Include medium and fine grids, and label this as a light check.	Grid checks should be layered and honest.

14 Limitations

Not included in Ticket 05	Reason
Boundary extraction	Planned for a later ticket; this report does not infer a free-boundary location.
Complementarity residual module	Planned diagnostic work after core validation layers.
Greeks	Delta and Gamma diagnostics are later work.
Stress maps, datasets, neural networks	These require a more complete validated solver pipeline.
No-dividend American call validation	Belongs to Ticket 06.
Dividend-paying American call validation	Belongs to a later call-validation ticket.
Full grid/domain sensitivity	The current medium-to-fine comparison is preliminary only.
External American reference prices	This ticket uses internal financial inequalities and solver metadata, not external benchmark tables.

15 Lessons Learned / Study Notes

The first lesson is that American put validation should not rely on one number. The value must be checked against payoff, against the European put, and against PSOR convergence metadata.

The second lesson is that obstacle gap is a validation diagnostic. It helps confirm that the American constraint is respected, but it should not be turned into a boundary claim without the later extraction logic.

The third lesson is that selected-spot tables are excellent for learning because they translate a surface into familiar moneyness points. They are not a replacement for full convergence and domain sensitivity studies.

The fourth lesson is that figures should support an argument. The Ticket 05 figure shows the relationship among American value, European value, and payoff. It is not decorative and does not label a free boundary.

16 Link to Ticket 06

Ticket 06 should validate the no-dividend American call theorem. Under the standard Black-Scholes assumptions with $q = 0$, a standard American call should match the corresponding European call. That check is different from Ticket 05: it tests whether the American exercise logic avoids inventing early exercise where theory says it should not occur.

17 Reproducibility Record

17.1 Files Created

Path	Role in Ticket 05
experiments/02_american_put_validation.py	Runs the American put validation cases and writes CSV/figure artifacts.
tests/test_american_put_validation.py	Focused unittest coverage for Ticket 05 artifacts, metrics, scope, and docstrings.
results/01_solver_validation/tables/ticket_05_american_put_validation.csv	Validation summary table.
results/01_solver_validation/tables/ticket_05_american_put_selected_spots.csv	Selected moneyness price comparison table.
results/01_solver_validation/figures/ticket_05_american_put_value_vs_payoff.png	American put value, payoff, and European put comparison figure.
reports/03_solver/tickets/ticket_05_american_put_validation.tex	Formal LaTeX study-report source.
reports/03_solver/tickets/ticket_05_american_put_validation.pdf	Formal PDF compiled from the LaTeX source.

17.2 Commands

Test command:

```
PYTHONPYCACHEPREFIX=/private/tmp/codex_pycache PYTHONPATH=src MPLCONFIGDIR=/private/tmp/matplotlib-cache python3 -m unittest discover -s tests
```

Experiment command:

```
PYTHONPYCACHEPREFIX=/private/tmp/codex_pycache PYTHONPATH=src MPLCONFIGDIR=/private/tmp/matplotlib-cache python3 experiments/02_american_put_validation.py
```

Compile command:

```
PYTHONPYCACHEPREFIX=/private/tmp/codex_pycache PYTHONPATH=src python3 -m compileall -q src tests experiments
```

LaTeX compile command:

```
HOME=/private/tmp FC_CACHEDIR=/private/tmp/fontconfig-cache latexmk -xelatex -interaction=nonstopmode -halt-on-error -outdir=/private/tmp/ticket05_latex reports/03_solver/tickets/ticket_05_american_put_validation.tex
```

PDF verification uses `pdftinfo` and page rendering with `pdftoppm`. Raw Git status is intentionally not included in this formal PDF report.

References

- Fischer Black and Myron Scholes. The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654, 1973. doi: 10.1086/260062.
- Michael J. Brennan and Eduardo S. Schwartz. The valuation of american put options. *The Journal of Finance*, 32(2):449, 1977. doi: 10.2307/2326779.

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Robert C. Merton. Theory of rational option pricing. *The Bell Journal of Economics and Management Science*, 4(1):141, 1973. doi: 10.2307/3003143.